

Remittance API Design Specification Document

Version 3.2

Nov, 2022

Contact Person

Name	Geoffrey Wambugu
Email	Geoffrey.kamau@equitybank.co.ke

Table of Contents

Table of Contents	2
Change History	3
1.0 Introduction.....	4
1.1 Transaction Process Flows	4
1.2 API Description.....	10
1.2.1 Remittance API	10
1.2.2 Status API	15
1.3 Response/Error codes	16
1.4 Code Samples.....	18
Referenced documents.....	21

Change History

Change Date	Version	Author	Change Description
2017/05/15	3.0	Andrew Mlamba	Add API Endpoint, Remove custom specs for Western Union.
2017/06/06	3.0	Andrew Mlamba	Add status enquiry method.
2021/04/01	3.1	Shadrack Mwagiru	- Added sender country tag - Added beneficiary name - Made sender name mandatory - Added 2 response codes (992 & 993) - Added sequence diagrams to include AML screening - Added AML screening API in the referenced documents section
2021/04/13	3.1.1	Geoffrey Wambugu	- -Added AML failure response code
2022/06/09	3.1.2	Shadrack Mwagiru	- Added -5 & 2080 response codes for telco specific error codes
2022/11/17	3.2	Shadrack Mwagiru	- Added senderDOB, SenderIDType, senderIDNumber, senderPhoneNumber, SenderCurrency, PurposeofRemittance fields in the transaction API

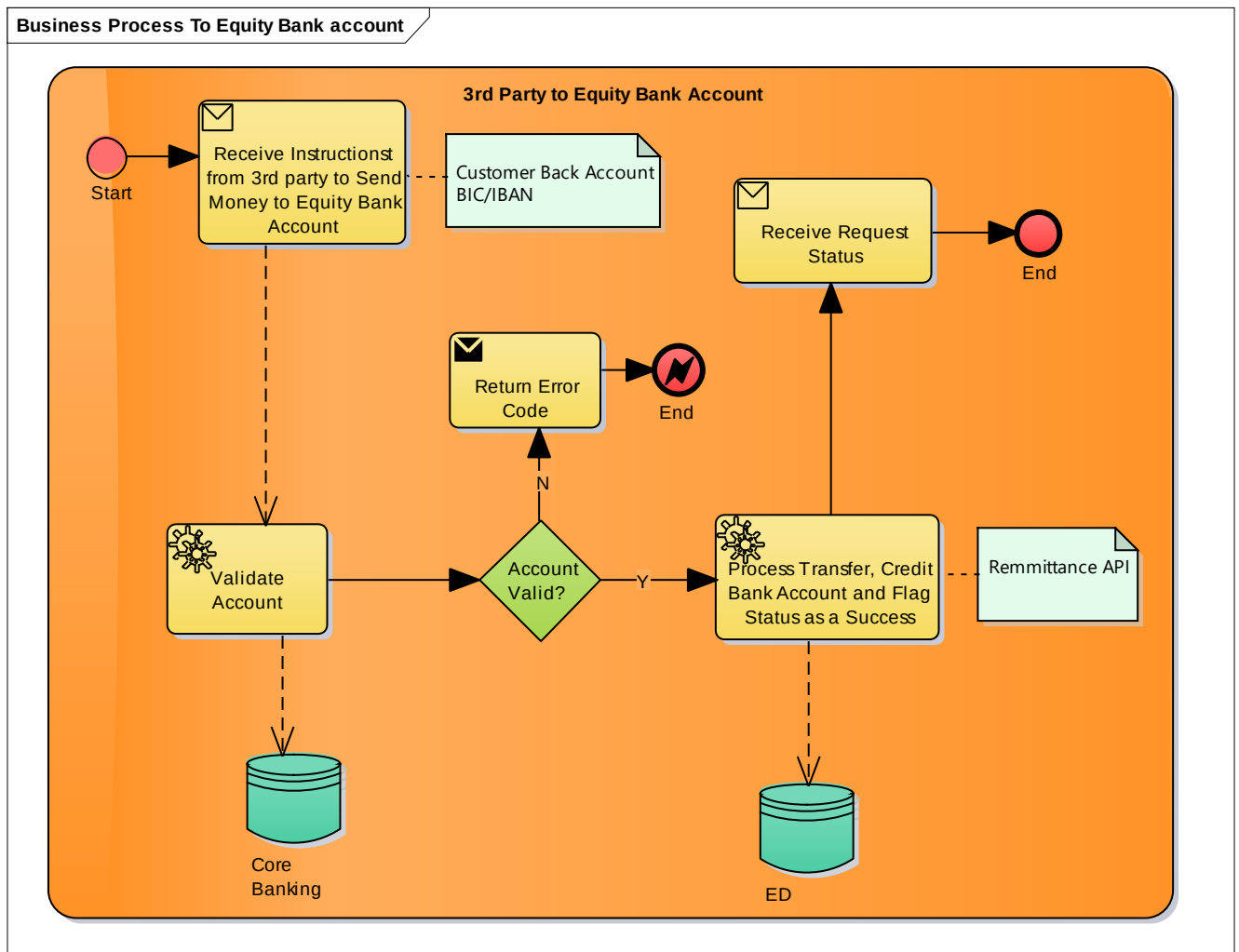
1.0 Introduction

Remittance API is a webservice API exposed to third party money transfers organisations to facilitate transfers of funds to:

- Equity Bank Customers Kenya
- Other Banks Customers in Kenya
- Mobile Wallets in Kenya
- Mobile Wallets in the Subsidiaries
- To Equity Bank Subsidiaries Countries (Rwanda, Uganda, South Sudan and Tanzania)

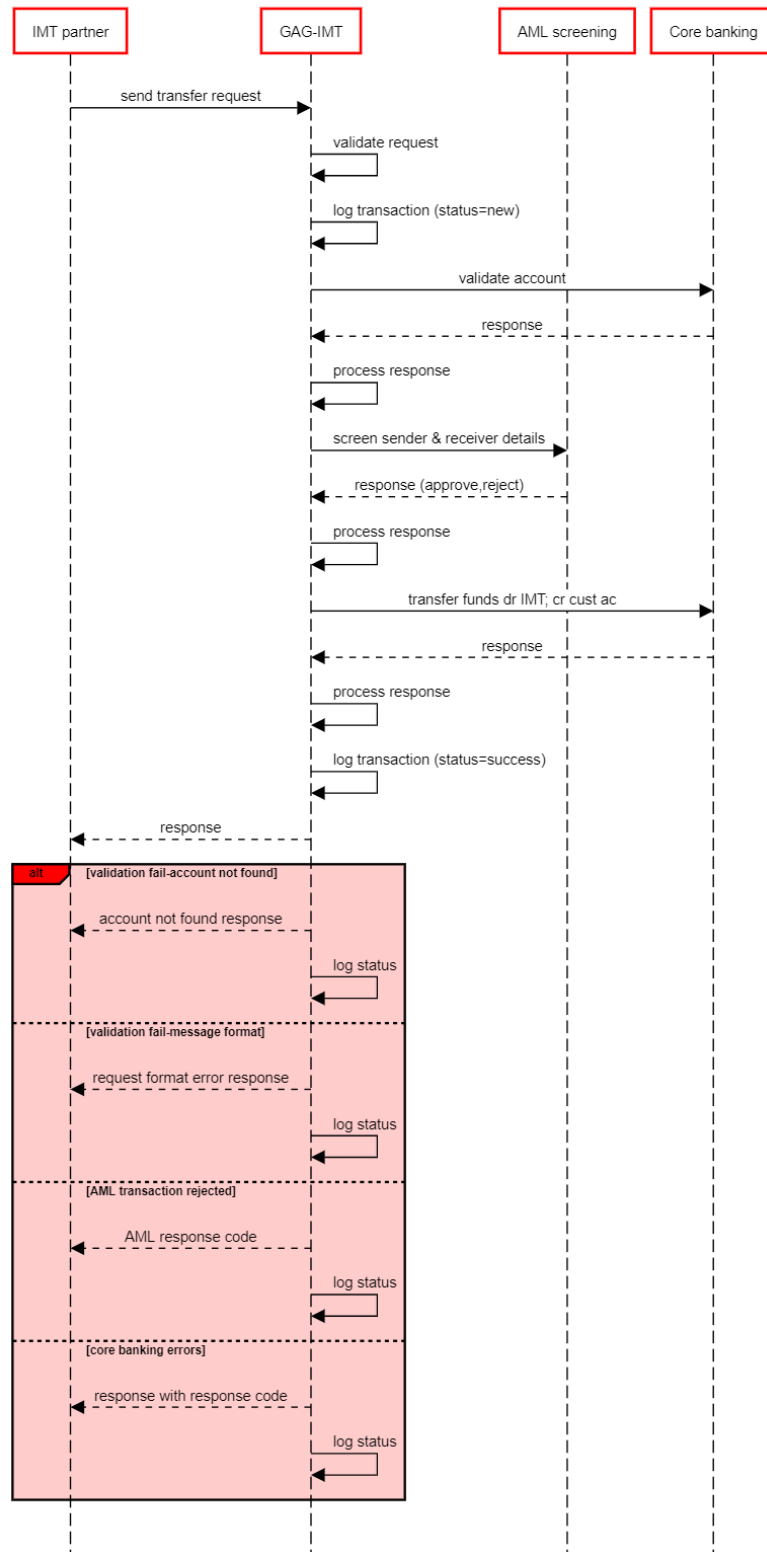
1.1 Transaction Process Flows

To Equity Bank



Sequence diagram to Equity bank

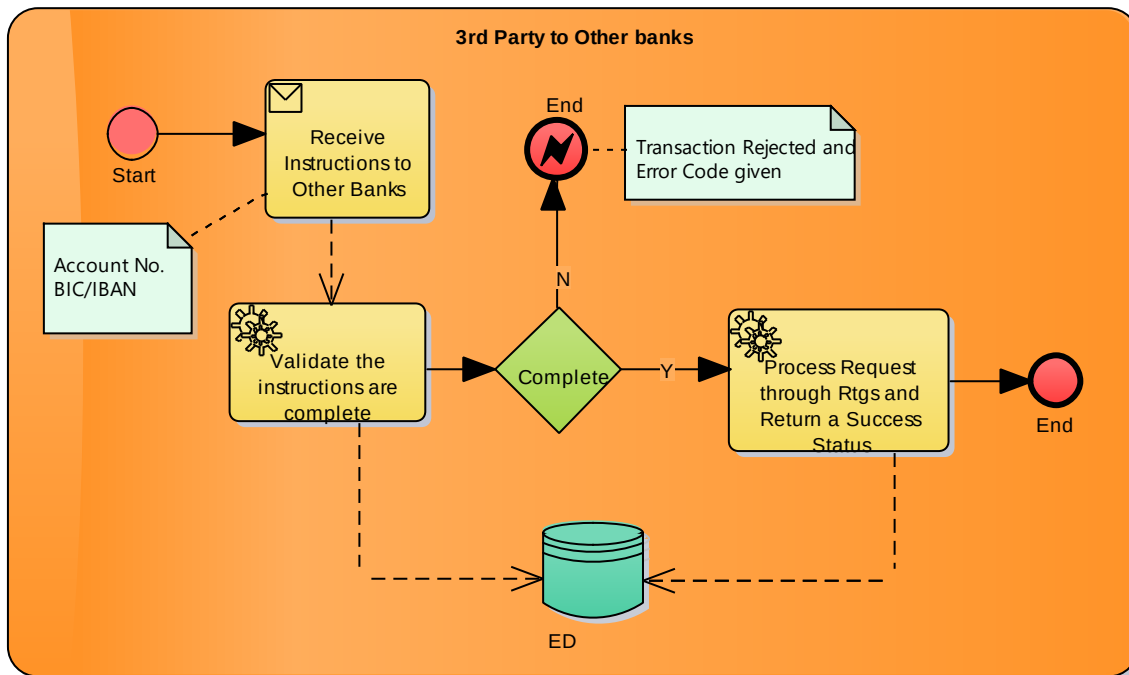
To Equity Bank



See referenced documents section for AML screening API

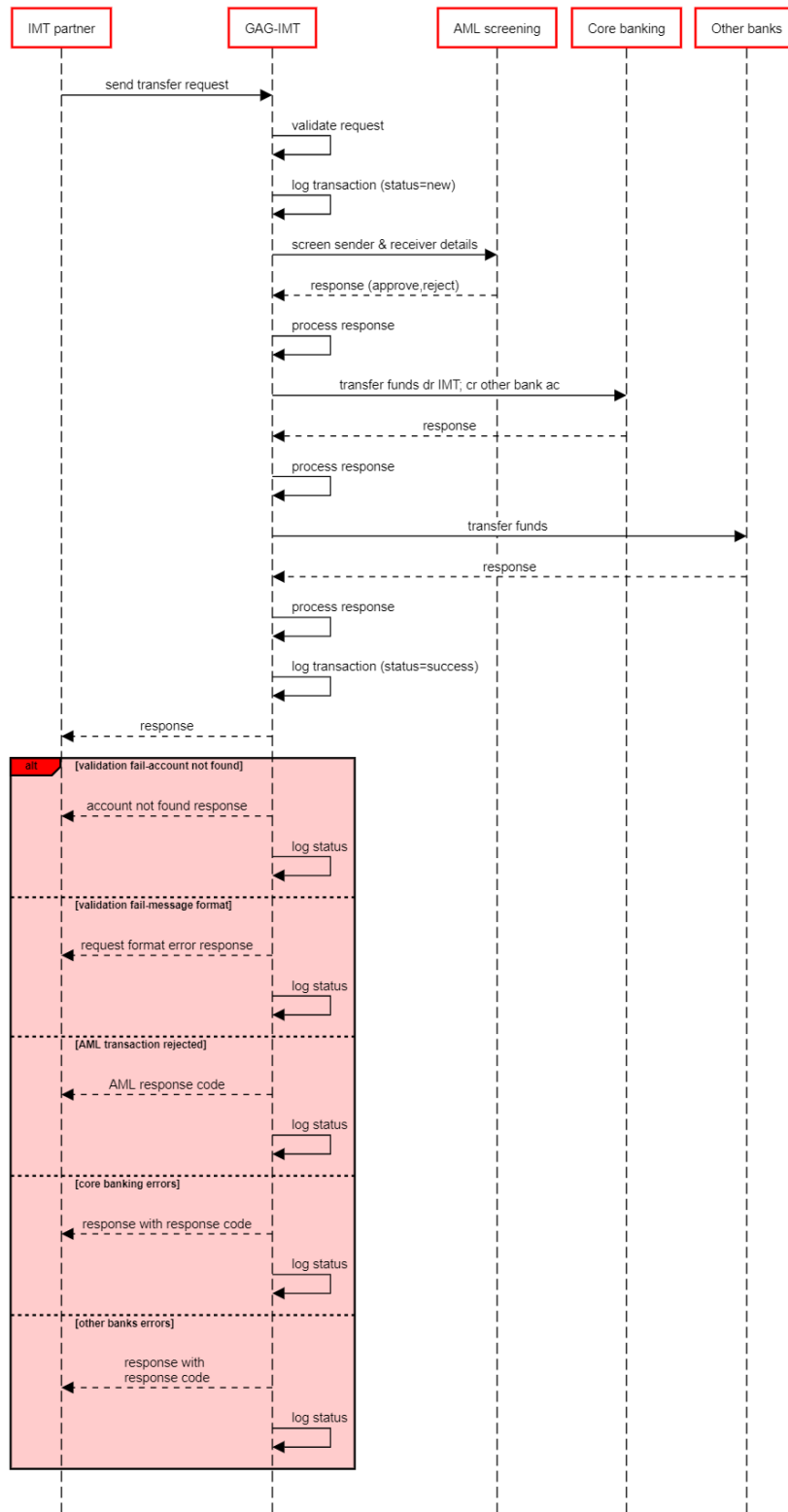
To Other Banks

Business Process To Other Banks



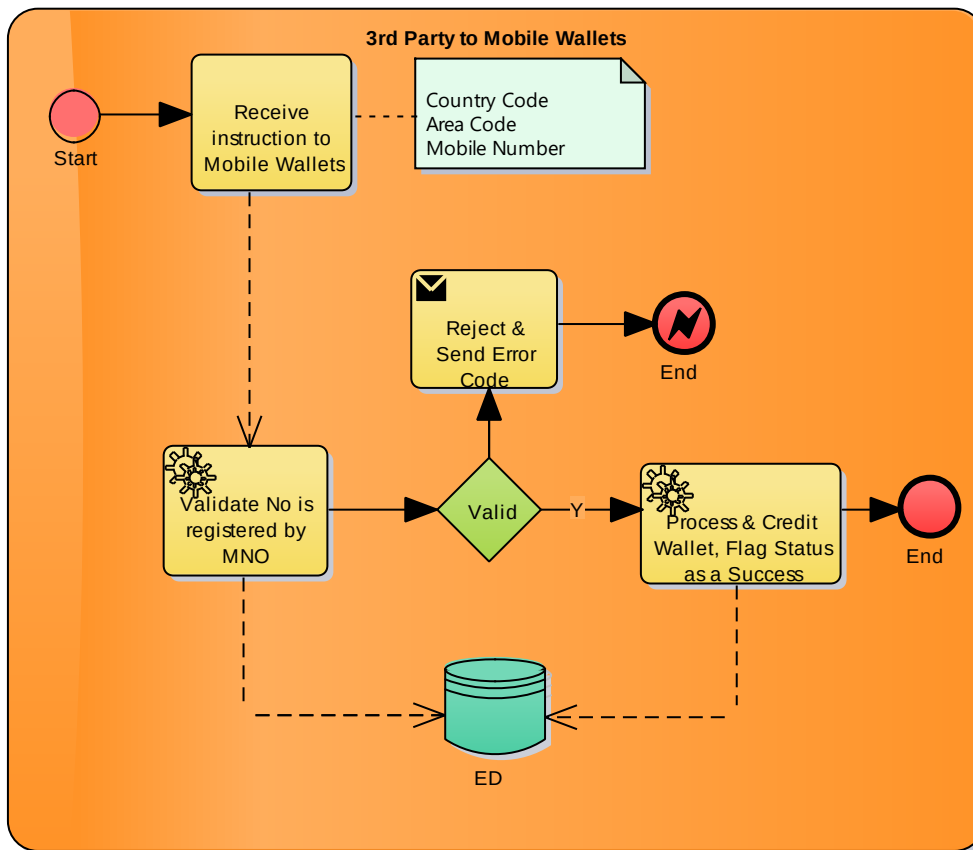
Sequence diagram to other banks

To other Banks

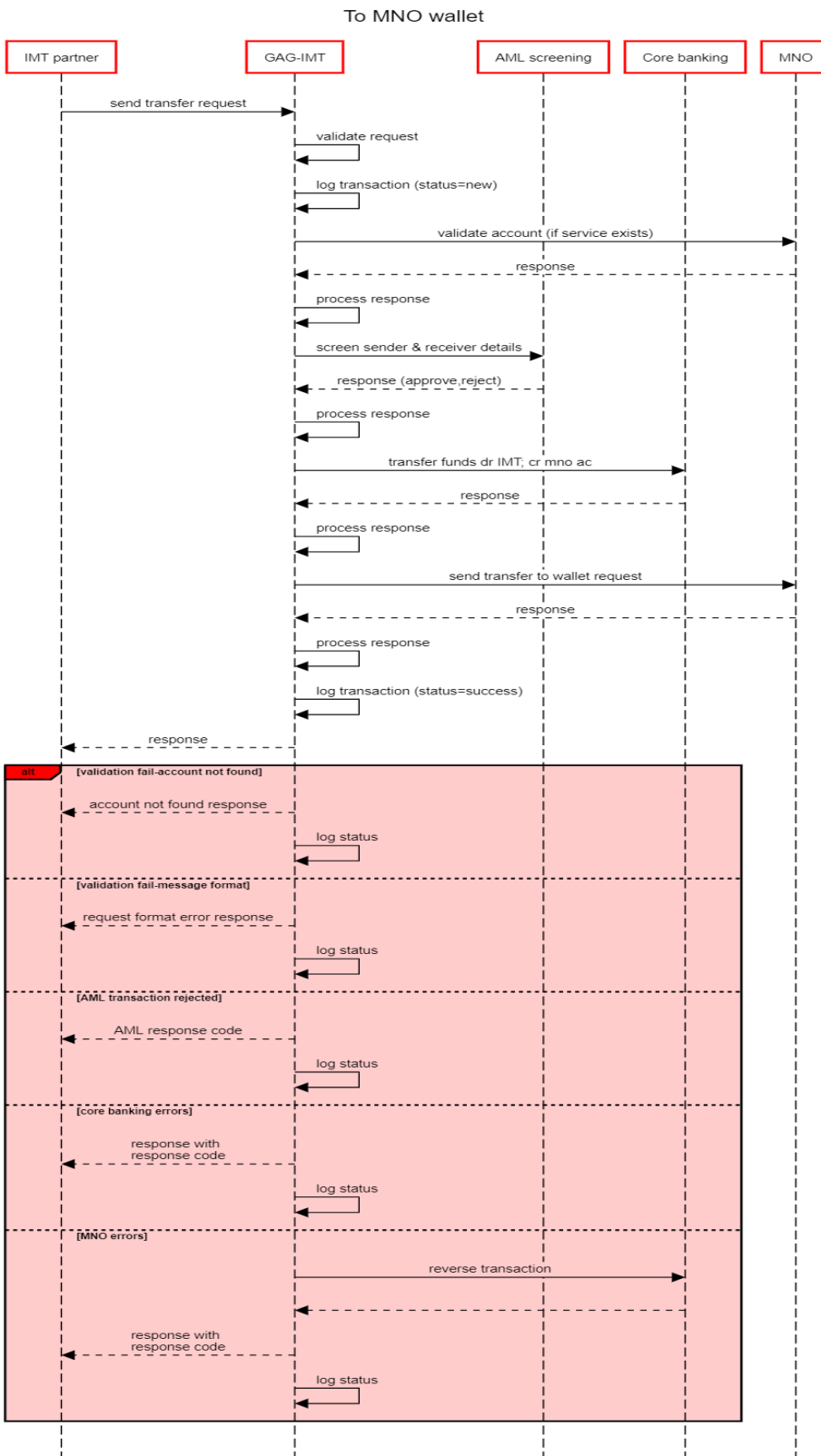


To Mobile Wallets

Business Process To Mobile Wallets



Sequence diagram to Mobile wallets



1.2 API Description

Endpoint: <https://wsuat.equitybankgroup.com/v1/transactionprocessor>

The API is SOAP webservice published on HTTPS and is accessible via a secure VPN or internet access. The requests and response are XML format.

The following are the fields description:

For internet access, IP or IP range whitelisting will be required at API gateway. For IMTs not on VPN or whose IPs keep changing (e.g those behind a CDN), they will be required to have mutual SSL with Equity bank.

IMTs will encrypt the payload using the public certificate shared with them.

IMTs will also sign the message using the key provided to them by Equity bank

Service	Operation	Dataservices
Remittance API	TransApply	Process Funds transfers
Status API	TranApplyRequestStatus	Get transaction Status

1.2.1 Remittance API

Request Field Mapping

Source Service Tag	Data Type	Length	Variable Length	Mand /Opt	Description Default/Expected Values
AccountNumber	String/text	13	Y	M	Beneficiary Bank Account Number
PaymentReference1	String/text	10	Y	M	Payment reference number e.g., Unique Transaction code/Reference. This field will be used to reconcile and batch proofing.
PaymentReference2	String/text	100	Y	M	Additional Payment reference e.g., Beneficiary Name or the Account Name

BeneficiaryName	String	100	Y	M	Beneficiary name
PaymentReference3	String/text	100	Y	M	Additional Payment reference e.g., Sender details or source transaction reference number
PaymentType	Lookup	50	Y	M	Static - The name of the remittance agent e.g. FUNTECH
Remarks	String/text	100	Y	O	The detail of the recipient of the funds i.e., the name of the beneficiary or the name of the account where money will be credited.
AgentID	Number	10	Y	M	A unique number to use to identify the paying account (to be shared)
Check	String/text	200	Y	M	This field is a checksum. The transaction acquiring organization/agent will sign this field with their private Key. First the following fields are concatenated. (SALT + (AccountNumber +BIC) or (Wallet +Mobile Number))+ paymentReference1+ paymentType+ agentID+ tranAmount+ currencyCode+messageID) in that order (Salt to be shared) (Mandatory). The resulting text is then signed with Private Key and Base64 encoded.(Mandatory)
Data	String/text	200		O	Not used for now (Future use)
MessageID	String/text	100		M	A unique number for every call to the web service
Fees	Float	-		M	Not used for now (Future use) default to Zero
InstCode	String/text	20		M	For Kenya use 54, Uganda 56, Tanzania 55, Rwanda 50 and South Sudan 11
senderCountryCode	String	2	N	M	Sender country code as defined by ISO 3166
SenderCurrency	String/text	20	N	M	Sender currency, conversation rate used and original amount (Values should be separated by a comma)

CurrencyCode	String/text	3	N	M	The ISO currency code of the transaction Amount. This is the receiver currency
CountryCode	String/text	2	N	M	Destination country code as defined by ISO 3166
BicCode	String/text	50	Y	O	Destination Bank Swift Bic code. For transactions to be remitted to other banks
MobileNumber	String/text	12	N	O	In case funds are remitted to a Telkom Wallet recipient Mobile number
WalletName	String/text	50	Y	O	An identifier to identify Telkom Wallet
BankCode	String/text	5		O	Not Used now
BranchCode	String/text	5		O	Not Used Now
SenderName	String/text	100	Y	M	A sender name is required for SMS Notification
SenderPhoneNumber	String/text	20	Y	M	Sender phone number
SenderDOB	String/text	10	N	M	Sender Date of Birth (yyyy-mm-dd). This can also have the value null
SenderNationality	String/text	20	N	M	Sender Nationality
SenderIDType	String/text	20	Y	M	Sender ID Type. Passport, National ID, Driver's License or Diplomatic Passport. This can also have the value null
SenderIDNumber	String/text	20	Y	M	Sender ID number. This can also have the value null
PurposeofRemittance	String/text	100	N	M	Purpose of remittance
TransAmount	float	-	Y	M	Transaction Amount in Receivers currency

Response Field Mapping

Source ServiceTag	Target Field	Data Type	Length	Description Default/Expected Values
StatusMessage	Status.Text	String	100	Success or Failure Message Success → OK {TranId} Error → ERROR {Errorcode}

Request XML

```
<?xml version="1.0" encoding="UTF-8"?>

<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">

  <S:Header>

  </S:Header>

  <S:Body>

    <ns0:tranApplyRequest xmlns:ns0="http://ws.equitybank/">

      <accountNumber>100019437XXXX</accountNumber>

      <paymentReference1>Bridge Educational Gaps Project</paymentReference1>

      <paymentReference2>323435445</paymentReference2>

      <beneficiaryName>Test beneficiary</beneficiaryName>

      <paymentReference3>sender@gmail.com </paymentReference3>

      <paymentType>WU</paymentType>

      <remarks>Geoffrey Doe</remarks>

      <agentID>5</agentID>

      <check>AB80uJ1h2G+SDAL6qwNsVW3HaYEI/5zvSuPhl+gDLITfZORxsLhK7rP2EBnt6jePpwWWDEahYHsmH9WTZHaNDizhUEK4FLiaH6qrgx06
y45zyZvTBKUAHFsox5V04XQOTVO2YQ3JRZYaQcgRCV+qmDUfQQmTqPs2rzly5pE1yTRVA/McF/b6d8yip8wQ5C9utq6l58sj2g7WuKeDjIVj4u
7C6xsJI+h/KZXUq348QpYV929wO76pYidJBjLRmJjevdDx7gZ23NxxnJIB+waBnUBNpsocFyCouqXyYDWWpCk8pYq/vc/FP1rlKgxLywD9jzFt4galZ
pM+fc51qw+Lg==</check>

      <fees>0.0</fees>

      <instCode>KE</instCode>

      <senderCountryCode>KE</senderCountryCode>
```

```

<tranAmount>72800.0</tranAmount>

<data>518227</data>

<messageID>518227</messageID>

<currencyCode>KES</currencyCode>

<countryCode>KE</countryCode>

<bicCode>EQBLKENA</bicCode>

<bankCode />

<branchCode />

<senderName>Ali, Yusuf</senderName>

<senderDOB>1970-01-09</senderDOB>

<senderPhoneNumber>+25483094</senderPhoneNumber>

<senderIDType>NationID</senderIDType>

<senderIDNumber>34343434</senderIDNumber>

</ns0:tranApplyRequest>

</S:Body>

</S:Envelope>

```

>

Response XML

```

<?xml version="1.0" encoding="UTF-8"?>

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Body>

    <ns2:tranApplyRequestResponse xmlns:ns2="http://ws.equitybank/">

      <return>OK|300776091932</return>

    </ns2:tranApplyRequestResponse>

```

```
</soap:Body>
```

```
</soap:Envelope>
```

1.2.2 Status API

Request Field Mapping

Source Service Tag	Data Type	Length	Variable Length	Mand/Opt	Description - Default/Expected Values
messageID	Text/String	20	Y	M	
agentID	Text/String	10	Y	M	
paymentReference1	Text/String	50	Y	M	
check	Text/String	200	Y	M	sign and encode: case paymentType optional: salt + MessageID + AgentID + PaymentReference1 case paymentType is supplied salt + MessageID + paymentType + PaymentReference1
paymentType	Text/String	100	Y	O	

Response Field Mapping

Source Service Tag	Data Type	Length	Variable Length	Mand/Opt	Description - Default/Expected Values
return	Text/String	100	Y	M	

Request XML

```
<?xml version="1.0" encoding="UTF-8"?>
<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">
  <S:Header>
  </S:Header>
  <S:Body>
    <ns0:tranApplyRequestStatus xmlns:ns0="http://ws.equitybank/">
      <messageID>518227</messageID>
```

```
<agentID>5</agentID>
<paymentReference1>Bridge Educational Gaps Project</paymentReference1>
```

```
<check>AB80uJ1h2G+SDAL6qwNsVW3HaYEI/5zvSuPhl+gDLITfzORxsLhK7rP2EBnt6jePpwWWDEahYHsmH9WTZHaNDizhUEK4
FLiaH6qrgx06y45zyZvTBKUAHFsox5V04XQOTVO2YQ3JRZYaQcgRCV+qmDUfQZQmTqPs2rzly5pE1yTRVA/McF/b6d8yip8wQ5C
9utq6l58sj2g7WuKeDjIVj4u7C6xsJI+h/KZXUq348QpYV929wO76pYidJBjLRmJjevdDx7gZ23NxxnJIB+waBnUBNpsocFyCouqXyYD
WWpCk8pYq/vc/FP1rlKgXLywD9jzFt4galZpM+fc51qw+Lg==</check>
  </ns0:tranApplyRequestStatus>
</S:Body>
</S:Envelope>
```

Response XML

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse xmlns:ns2="http://ws.equitybank/">
      <return>OK|300776091932|PAID SUCCESSFUL|MSGID=256454</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

Response XML – Telco specific errors

```
<soap:Envelope
  xmlns:soap=http://schemas.xmlsoap.org/soap/envelope/>
  <soap:Body>
    <ns2:tranApplyRequestStatusResponse
      xmlns:ns2=http://ws.equitybank/>
      <return>ERROR|119|2080</return>
    </ns2:tranApplyRequestStatusResponse>
  </soap:Body>
</soap:Envelope>

<soap:Envelope
  xmlns:soap=http://schemas.xmlsoap.org/soap/envelope/>
  <soap:Body>
    <ns2:tranApplyRequestStatusResponse
      xmlns:ns2=http://ws.equitybank/>
      <return>ERROR|119|2040</return>
    </ns2:tranApplyRequestStatusResponse>
  </soap:Body>
</soap:Envelope>
```

1.3 Response/Error codes

1.3.1 Status error code and description

ERROR_CODE	ERROR_MESSAGE
------------	---------------

906	Cut over (end of day) in progress
911	Host timed out
116	Insufficient Funds
114	Invalid Account Number
902	Invalid function code
115	Invalid processing code
904	Message incorrectly formatted
908	Processing host cannot be reached
907	Processing host is down
909	System Malfunction
000	Transaction Approved
119	Transaction Not Permitted
180	Transfer limit exceeded
121	Withdrawal limit exceeded
77	Payment Transaction Not found
-1	Validation: System error
-2	Validation: failed Missing Parameters
-3	Transaction Failed- Core banking system general error
-4	Transaction Failed- Failed to reach third party system
-5	Validation:Telco validation failed
-7	Validation:Transaction Duplicate
-9	Invalid or Unrecognized Wallet code
-10	Validation Deposit Limits exceeded
-12	AML Failure
-99	Validation: Transaction Integrity Failed
992	Transaction rejected because of AML reasons
993	Transaction Held for AML investigation

1.3.2 Additional Wallet Error Codes on Status API

WALLET NAME	ERROR CODE	ERROR DESCRIPTION
Mpesa	2	Limit rule: less than the minimum transaction amount

Mpesa	7	Receiver mobile number is in invalid
Mpesa	14	Receiver mobile number is in an invalid state
Mpesa	8	Would exceed the maximum balance
Mpesa	2040	Receiver mobile number is unregistered in MPesa
Mpesa	2080	Maximum limit exceeded

1.4 Code Samples

Java

```
package equitybank.ws;
```

```
import java.io.File;
import java.net.MalformedURLException;
import java.net.URL;
import javax.xml.namespace.QName;
import javax.jws.WebMethod;
import javax.jws.WebParam;
import javax.jws.WebResult;
import javax.jws.WebService;
import javax.xml.bind.annotation.XmlSeeAlso;
import javax.xml.ws.RequestWrapper;
```

```
import javax.xml.ws.ResponseWrapper;

public final class TransApplyClient {

    private static final QName SERVICE_NAME = new QName("http://ws.equitybank/", "TransApply");

    private TransApply_TransApplyImpPort_Client() {

    }

    public static void main(String args[]) throws java.lang.Exception {
        URL wsdlURL = TransApply_Service.WSDL_LOCATION;
        if (args.length > 0 && args[0] != null && !"".equals(args[0])) {
            File wsdlFile = new File(args[0]);
            try {
                if (wsdlFile.exists()) {
                    wsdlURL = wsdlFile.toURI().toURL();
                } else {
                    wsdlURL = new URL(args[0]);
                }
            } catch (MalformedURLException e) {
                e.printStackTrace();
            }
        }

        TransApply_Service ss = new TransApply_Service(wsdlURL, SERVICE_NAME);
        TransApply port = ss.getTransApplyImpPort();

        {
            System.out.println("Invoking tranApplyRequest...");
            java.lang.String _tranApplyRequest_accountNumber = "";
            java.lang.String _tranApplyRequest_paymentReference1 = "";
            java.lang.String _tranApplyRequest_paymentReference2 = "";
            java.lang.String _tranApplyRequest_paymentReference3 = "";
            java.lang.String _tranApplyRequest_paymentType = "";
            java.lang.String _tranApplyRequest_remarks = "";
            java.lang.String _tranApplyRequest_agentID = "";

            java.math.BigDecimal _tranApplyRequest_fees = new java.math.BigDecimal("0");
            java.lang.String _tranApplyRequest_instCode = "";
            java.math.BigDecimal _tranApplyRequest_tranAmount = new java.math.BigDecimal("0");
            java.lang.String paramdata = "";
            java.lang.String parammessageID = "";
            java.lang.String paramcurrencyCode = "";
            java.lang.String paramcountryCode = "";
            java.lang.String parambicCode = "";
        }
    }
}
```

```

java.lang.String parammobileNumber = "";
java.lang.String paramwalletName = "";
java.lang.String parambankCode = "";
java.lang.String parambranchCode = "";
java.lang.String paramsenderName = "";
java.lang.String paramcallBackUrl = "";

String theXsumFields = "";
    if ((paramBicCode != null) && (!paramBicCode.isEmpty())) {
        theXsumFields = vfxSalt + paramAccountNumber + paramBicCode +
paramPaymentReference1
        + paramPaymentType + paramAgentID + paramTranAmount.setScale(2,
RoundingMode.CEILING) + paramCurrencyCode + paramMessageID;
    } else if ((paramWalletName != null) && (!paramWalletName.isEmpty())) {
        paramWalletName = paramWalletName.toUpperCase();
        theXsumFields = vfxSalt + paramWalletName + paramMobileNumber +
paramPaymentReference1
        + paramPaymentType + paramAgentID + paramTranAmount.setScale(2,
RoundingMode.CEILING) + paramCurrencyCode + paramMessageID;
    }

    Base64 base64 = new Base64();
    java.lang.String paramcheck = base64.encode(signData(theXsumFields.getBytes()));
    java.lang.String param_return = port.tranApplyRequest(paramaccountNumber,
parampaymentReference1, parampaymentReference2, parampaymentReference3, parampaymentType,
paramremarks, paramagentID, paramcheck, paramfees, paraminstCode, paramtranAmount, paramdata,
parammessageID, paramcurrencyCode, paramcountryCode, parambicCode, parammobileNumber,
paramwalletName, parambankCode, parambranchCode, paramsenderName, paramcallBackUrl);
    System.out.println("tranApplyRequest.result=" + param_return);

}

System.exit(0);
}

    public static byte[] signData(byte[] data) {
    try {
        Security.addProvider(new BouncyCastleProvider());
        KeyStore ks = KeyStore.getInstance(KEYSTORE_INSTANCE);
        ks.load(new FileInputStream(KEYSTORE_FILE), KEYSTORE_PWD.toCharArray());
        Key key = ks.getKey(KEYSTORE_ALIAS, KEYSTORE_PWD.toCharArray());
        //Sign
        PrivateKey privKey = (PrivateKey) key;
        Signature signature = Signature.getInstance("SHA256withRSA", "BC");
        signature.initSign(privKey);
        signature.update(data);
    }
}

```

```

        return (signature.sign());
    } catch (KeyStoreException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (FileNotFoundException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (IOException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (NoSuchAlgorithmException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (CertificateException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (UnrecoverableKeyException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (NoSuchProviderException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (InvalidKeyException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    } catch (SignatureException ex) {
        Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
    }
    return null;
}
}

```

Referenced documents

Document	Description
 IMT - API - Overview.pdf	AML API used for screening against names, countries, amounts for IMTs